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Prop clase ck velocidad convergencia uniforme fourier

Proposición 1. *Sea* $k \in \mathbb{N}$ *y* $f \in \mathcal{C}^k\left(\left[-1/2, 1/2\right]\right)$ 1*-periódica*

$$\implies \exists C_k < \infty : \forall N \in \mathbb{N} : \|S_N f - f\|_\infty \leqslant \frac{C_k}{N^{k-1/2}}.$$

***Demostración*:** Por el `lem-serie-fourier-derivada/`lema 1, $\forall n \in \mathbb{Z} \setminus \{0\} : \widehat{f^{(k)}}(n) = (2\pi i n)^k \widehat{f}(n)$. Además, como $f \in \mathcal{C}^k\left(\left[-1/2, 1/2\right]\right) \subset \mathcal{L}^2\left(\left[-1/2, 1/2\right]\right)$, por Plancherel, tenemos que

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \left|f^{(k)}(x)\right|^2 \mathrm{d}x = \sum_{n\in\mathbb{Z}} \left|\widehat{f^{(k)}}(n)\right|^2 = \sum_{n\in\mathbb{Z}} 4^k \pi^{2k} |n|^{2k} \left|\widehat{f}(n)\right|^2. \tag{1}$$

Por otro lado, por la `prop-criterio-dini/`proposición 1, $\sum_{n\in\mathbb{Z}} \widehat{f}(n) e^{2\pi i n x} = \lim_{N\to\infty} S_N f(x) = f(x)$. Entonces,

$$\begin{aligned}
\|S_N f - f\|_\infty &= \left\|\sum_{|n|>N} \widehat{f}(n) e^{2\pi i n x}\right\|_\infty \leqslant \sum_{|n|>N} \left|\widehat{f}(n)\right| = \sum_{|n|>N} \left|\widehat{f}(n)\right| |n|^k \cdot \frac{1}{|n|^k} \\
&\overset{\substack{\text{C-S} \\ \downarrow}}{\leqslant} \left(\sum_{|n|>N} \left|\widehat{f}(n)\right|^2 |n|^{2k}\right)^{1/2} \left(\sum_{|n|>N} \frac{1}{|n|^{2k}}\right)^{1/2}
\end{aligned}$$

Ahora, usando la ecuación (1), tenemos que

$$\begin{aligned}
\|S_N f - f\|_\infty &\leqslant \frac{1}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \left|f^{(k)}(x)\right|^2 \mathrm{d}x\right)^{1/2} \left(2 \sum_{n=N+1}^{\infty} \frac{1}{n^{2k}}\right)^{1/2} \\
&\leqslant \frac{\sqrt{2}}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \left|f^{(k)}(x)\right|^2 \mathrm{d}x\right)^{1/2} \left(\int_N^\infty \frac{1}{x^{2k}} \mathrm{d}x\right)^{1/2} \\
&= \frac{\sqrt{2}}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \left|f^{(k)}(x)\right|^2 \mathrm{d}x\right)^{1/2} \left(\frac{1}{(2k-1)N^{2k-1}}\right)^{1/2} = \frac{C_k}{N^{k-1/2}},
\end{aligned}$$

donde $C_k = \dfrac{\sqrt{2}}{2^k \pi^k \sqrt{2k-1}} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \left|f^{(k)}(x)\right|^2 \mathrm{d}x\right)^{1/2} < \infty.$ ■